

Dr. Chilperic Armel Foko Kuate



Multiscale modelling • Scientific software • Data analysis

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Research website | [GitHub](#) | ORCID: 0000-0002-0140-7588

Profile

Computational scientist with a PhD in quantitative and theoretical biology and research experience in mechanistic modelling, nonlinear dynamics and scientific software. I develop mathematical models and computational workflows that connect assumptions, simulation, observations and interpretation. Target roles: computational scientist, modelling scientist and research software developer in research-intensive teams.

Capabilities

Model & compute	ODE/DAE systems, enzyme kinetics, stochastic and population models, steady-state and stability analysis.
Analyse & infer	Parameter fitting, sensitivity, uncertainty, constrained and derivative-free optimisation; statistical data analysis.
Build & reproduce	Python, NumPy/SciPy, SALib, CasADi/IPOPT, CMA-ES; Git, Linux, Jupyter, testing and documentation. Julia, R and MATLAB experience; JavaScript scientific interfaces.

Experience

Researcher • Computational Cell Biology

Oct 2026–present

Heinrich Heine University Düsseldorf • CCB

Resumed a research appointment on 1 October 2026, continuing computational-biology and mechanistic-modelling work.

Independent research & FokoLab development

Nov 2025–Sep 2026

Düsseldorf, Germany

Built browser-based scientific tools and maintained modelling repositories; continued research writing and AI/ML training.

Postdoctoral researcher

Nov 2022–Oct 2025

CEPLAS / CCB, Heinrich Heine University Düsseldorf

- Integrated biochemical, thermal and hydraulic processes in plant-adaptation models; explored environmental response and physiological trade-offs.
- Developed simulation, optimisation, parameter-exploration and sensitivity workflows, with student supervision and interdisciplinary collaboration.

Doctoral researcher • PoLiMeR ITN

Aug 2019–Aug 2022

Quantitative and Theoretical Biology, Heinrich Heine University Düsseldorf

- Developed kinetic models of hepatic fatty-acid metabolism and synthesis; analysed equilibria, stability and parameter-dependent switching.
- Contributed to publications on enzyme-kinetic modelling and differentiable constraint-based metabolic models.

Research fellow

Nov 2017–Mar 2019

AIMS · Cameroon and Rwanda

Stochastic and deterministic T-cell modelling; scientific teaching and computational methods.

Selected technical work**FokoLab · scientific application**

Connected laboratories for simulation, optimisation, sensitivity, fitting, statistics and mechanistic visualisation. Structured custom models, source attribution, saved experiments and exportable results.

[Website](#) · [Repository](#)**Plant–environment modelling**

A biochemical–thermal–hydraulic framework for photosynthetic adaptation, implemented as reproducible computational experiments. Collaboration includes student research and experimental partners.

[Source](#)**Metabolic and cell-population models**

Fatty-acid kinetics, nonlinear metabolic behaviour, branching populations and division-tracking models; mathematical formulation paired with numerical analysis.

[Research repositories](#)**Education****PhD · Quantitative and Theoretical Biology**

2019–2023

Heinrich Heine University Düsseldorf, Germany. Completed 7 March 2023.

Dissertation: *Dynamic computational modeling of fatty acid de novo synthesis in the liver and qualitative analysis of fatty acid metabolism*. Supervision: Oliver Ebenhöf; co-supervision and collaboration: Adélaïde Raguin.

MSc · Mathematics and Fundamental Applications

2017–2019

Modelling specialisation · University of Yaoundé I, Cameroon.

MSc · Mathematical Sciences

2015–2016

AIMS Ghana / University of Cape Coast, Ghana.

BSc · Mathematics

2009–2012

University of Douala, Cameroon.

Baccalauréat

2009

Collège La Confiance, Bafoussam, Cameroon.

Selected publications

Foko Kuate, Ebenhöf, Bakker & Raguin (2023). Enzyme-kinetic data for animal fatty-acid synthesis. *Bioscience Reports*. 10.1042/BSR20222496.

Wilken et al., including Foko Kuate (2022). Differentiable constraint-based metabolic models. *Metabolic Engineering* 74, 72–82. 10.1016/j.ymben.2022.09.002.

Teaching, languages & development

Supervision of two MSc and two BSc projects; 200+ teaching/tutorial hours. French: native; English: proficient; German: B1/B1+, B2 study ongoing. AI engineering and machine-learning training complement established mechanistic-modelling expertise.